

To: The Editors, *Quantum Machine Intelligence*
Re: Submission of original research article
Topic: Quantum Computing for AI

Dear Editors,

We are pleased to submit our manuscript, “**How Quantum Is the Advantage? A Fair, Calibration- and Noise-Aware Benchmark and Attribution Audit of Quantum Machine Learning for Network Intrusion Detection,**” for consideration as an original research article in *Quantum Machine Intelligence*.

The question. Applied quantum machine learning reports near-perfect accuracies across many application domains, network intrusion detection among them, with figures in the 97 to 99.9% range. The field’s more careful studies have begun to question those numbers, showing that well-tuned classical models remain competitive on tabular data and that apparent quantum gains may be artefacts of the classical dimensionality reduction and implicit regularisation that every hybrid pipeline contains. What the literature lacks is not another accuracy record but a *method* for deciding, for a given result, how much of the measured advantage is actually attributable to the quantum component. That is what this paper supplies.

What we contribute. The methodological core is a **quantum-attribution audit**: for every quantum model we construct a parameter-matched classical control sharing the identical classical front-end; for every quantum kernel we construct a random-feature kernel as its explicit classical surrogate; and we run a regularisation sweep to test whether comparable regularisation closes any observed gap. Differences are reported per variant and metric with significance testing and Benjamini-Hochberg control of the false-discovery rate across the whole comparison family. The audit is general: it applies to any claim of quantum advantage in a hybrid learning pipeline, and we demonstrate it on a benchmark of two quantum model families (hybrid variational circuits and quantum-kernel SVMs) against five honestly-tuned classical baselines, across four standard datasets, under one leakage-controlled protocol with an equal-budget feature view, imbalance- and calibration-aware metrics, and a simulated NISQ noise sweep.

What we find. Tuned classical models match or exceed the quantum models on aggregate detection on every dataset, and the audit attributes this to classical preprocessing and regularisation rather than to quantum effects. Two advantages survive FDR control. The quantum-kernel SVM significantly out-ranks its direct classical surrogate on AUPRC ($q = 0.011$) and ROC-AUC ($q = 0.018$); because the random-feature kernel is the explicit classical analogue of the quantum feature map, this is the sharpest isolation of a genuinely quantum contribution in the study, and it is the one result that also survives the more conservative family-wise Holm correction. Separately, a small four-qubit hybrid out-detects the best classical baseline at the 1% false-positive operating point on a distribution-shifted task ($p = 0.005$, $q = 0.030$); we report plainly that this one does not survive Holm.

Fit for the journal. The work sits in the journal’s Quantum Computing for AI perspective and speaks directly to its stated emphasis on benchmarking and empirical evaluation. It is addressed to readers who want to know whether reported quantum advantages hold up under matched classical controls, which is a question for this readership rather than for an applications venue. We also believe the audit is reusable well beyond intrusion detection.

Limitations we state up front. All quantum results are simulated; we have no QPU access and treat the simulated numbers as an optimistic bound. Our seed budget is small and uneven (five seeds for the classical baselines on one dataset, three elsewhere and for all quantum runs, two for the noise sweep), which we report explicitly and which is why the operating-point claims

rest on a paired bootstrap rather than on seed-level tests. The attribution audit is decomposed on the most discriminative dataset. We prefer to surface these limits than have them inferred.

Reproducibility. The code, configurations, fixed seeds, the scripts that reconstruct our exact splits, and the per-run result records are released publicly; every reported number is regenerable from the released harness.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have approved the submission and declare no competing interests.

Thank you for your consideration.

Sincerely,

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